Jyun-Ru (Jin) Huang

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Education

Master of Science in Business Analytics Aug 2024 – Jan 2026
Boston University, Questrom School of Business (GPA: 3.58) *Boston, MA*

Bachelor of Arts in Economics, Minor in Political Science Aug 2014 – Jan 2019
National Taiwan University *Taipei, Taiwan*

Work Experience

CTBC Bank Jul 2020 – May 2023
Retail Credit Risk Analyst *Taipei, Taiwan*

- Developed Loss Given Default (LGD) forecasting models for mortgage loans using SAS and SQL, streamlining the process by narrowing 16,000 predictors to 10 key variables through feature engineering, improving model discrimination by 18% (Gini coefficient).
- Led research and modeling of typhoon flood impact on mortgage collateral by analyzing meteorological open data with ArcGIS, resulting in a patented geographic risk model in Taiwan.
- Led three risk analysis projects across mortgage, personal loans, and credit cards; one project resulted in a proactive credit card account closure policy that was successfully implemented.

Taipei Fubon Commercial Bank Jul 2019 – Jun 2020
Institutional Credit Risk Analyst *Taipei, Taiwan*

- Conducted credit risk analysis on 8+ large corporate borrowers with USD 10M+ exposure per case, evaluating financial metrics and conducting macroeconomic stress testing to support credit approval decisions.
- Managed end-to-end credit assessment processes including internal credit rating evaluations, risk case documentation, and credit approval workflow coordination.
- Enhanced Excel VBA financial forecasting models to expand applicability across companies of various sizes and industries, improving credit analysis efficiency.

E.Sun Commercial Bank Jul 2018 – Aug 2018
Credit Card Marketing Intern *Taipei, Taiwan*

- Supported bank-wide customer marketing list query requests using SQL to inform marketing campaign strategies.
- Conducted credit card marketing research using Tableau dashboards.
- Proposed a redesigned mobile banking credit card dashboard UI, which was adopted after my internship.

Project Experience

Predicting IMDb Movie Ratings - A Multimodal Deep Learning Framework Jun 2025 – Aug 2025
BA890: Analytics Practicum (Research Project), Boston University

- Consolidated 4,800+ films across nearly 40 years from three film databases and built a multi-page interactive Tableau dashboard with dimensional filters, trend lines, and distribution analysis.
- Extended EDA findings on studio release strategies into a multimodal deep learning framework built in PyTorch and trained on NVIDIA A100 GPU, integrating unstructured features from film synopses (text) and movie posters (image).
- Achieved an average prediction error within ±0.6 IMDb rating points (0–10 scale) on the validation set, a 30% reduction compared to traditional ML baselines using only structured features (budget, genre, runtime).

Real-Time ICU Demand Forecasting for Hospital Capacity Planning Sep 2025 – Dec 2025

BA878: Machine Learning in Healthcare, Boston University

- Built a comprehensive ICU bed demand forecasting framework using MIMIC-IV data to support hospital resource planning.
- Developed two independent XGBoost-based predictive models to estimate ICU inflow from the emergency department within 12 hours and 72-hour readmission risk after ICU discharge.
- AUC = 0.96 for ICU inflow and AUC = 0.72 for the readmission model, exceeding performance reported in prior literature.

U.S. Military Base Slot Machine Revenue Explorer Sep 2025 – Dec 2025

DS701: Tools for Data Science, Boston University

- Supported MuckRock, a nonprofit investigative journalism organization, by cleaning and analyzing publicly released U.S. military slot machine revenue, location, and related datasets.
- Built a layout-aware, rule-based data extraction pipeline to parse borderless PDF tables into clean, analysis-ready datasets using Python.
- Parsed and structured 203 pages of borderless, irregular PDF tables in under 3 minutes, reducing manual data processing time from days to minutes.

Case Competition Experience

Humana-Mays 2024 Healthcare Analytics Case Competition Sep 2024 – Nov 2024

Boston University — Placed Top 50 (Round 2), AUC Top 10 among 200+ teams

- Identified "unengaged" Humana members lacking a preventive PCP visit and proposed data-driven strategies to increase visit rates.
- Engineered and selected features across 14 datasets (500+ variables), building an optimized XGBoost predictive model.

Skills

- **Programming:** Python, SQL, Excel VBA
- **Machine Learning / Deep Learning:** PyTorch, TensorFlow
- **Big Data:** PySpark, BigQuery
- **Cloud:** Google Cloud Platform (GCP)
- **Data Visualization:** Tableau
- **Geospatial Analysis:** Esri ArcGIS
- **Statistics:** SAS (SAS Base Programming Certified)